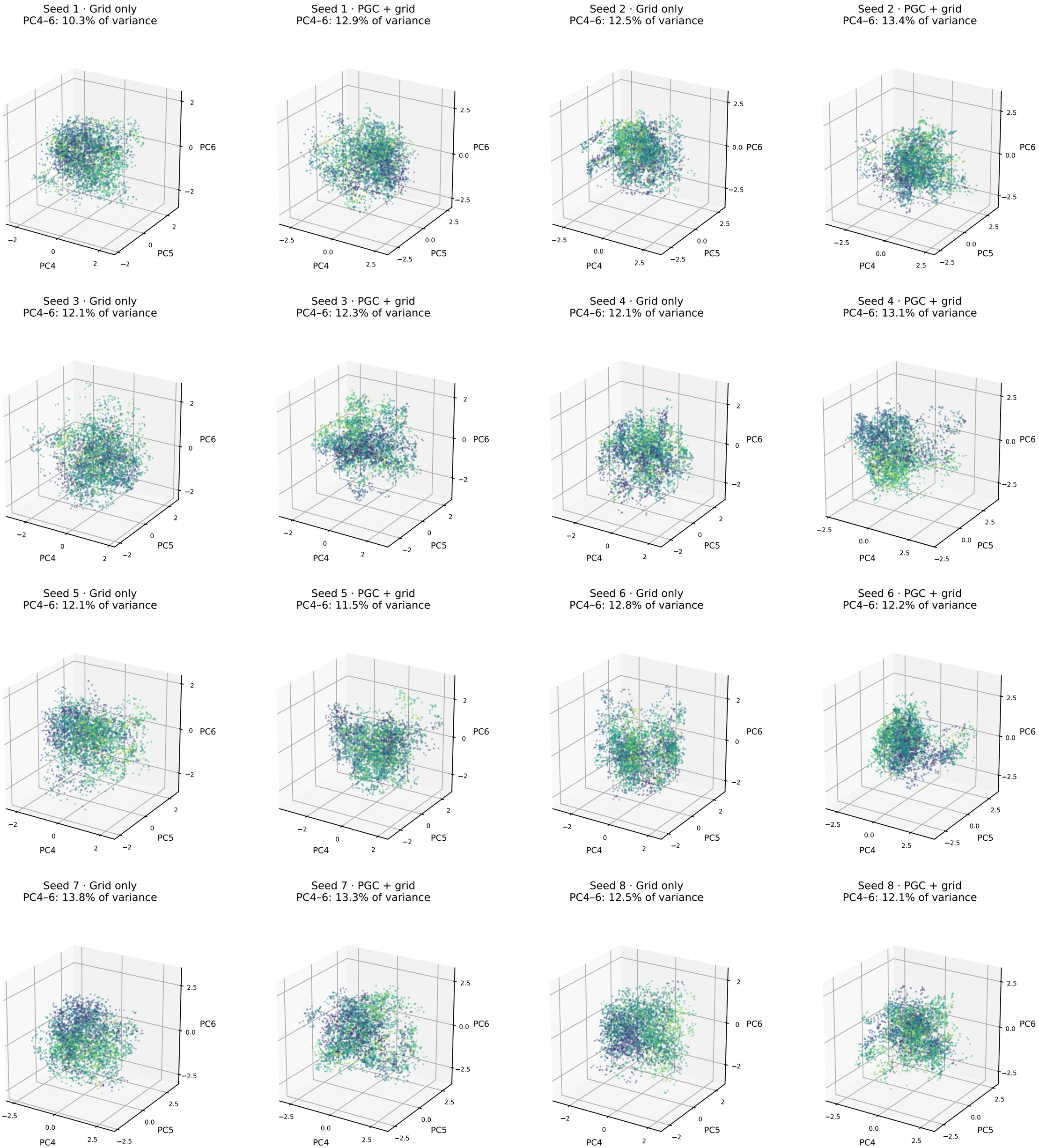
All eight seed pairs · frozen LSTMs at 1B frames · PCs 4–6
3,393 matched states: varying movement, facing, and remembered goals



Independent centered PCA per model; no whitening or UMAP. Equal scale within each 3-D panel; axes are not aligned across models. Shape alone is not a gridness or torus test.